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# Data Envelopment Analysis and hyperbolic efficiency measures: Extending applications and possibilities for between-group comparisons

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ABSTRACT

Data Envelopment Analysis (DEA) is a widely used tool to estimate relative efficiencies. However, this methodology encounters issues when between-group comparisons with variable returns of scale (VRS) are of interest. In this specification efficiency scores can be undefined when the projection of the efficiency estimation of an observation from one group does not have an intersection with the frontier of the other group. To address this issue, a hyperbolic efficiency estimation can be used. Up to now, the literature lacks consideration of weight restrictions and non-discretionary variables in the hyperbolic DEA method. Weight restrictions allow redefining the production possibility set (PPS), while non-discretionary variables adjust the model's orientation. Considering these factors, it is possible to incorporate prior knowledge, substitution effects, and increase discriminatory power. This paper introduces the respective mathematical formulations and the practical implementation in statistical software. The estimation methods have been combined in an R-package called *hyperbolicDEA* in order to facilitate the application of hyperbolic DEA with weight restrictions, non-discretionary variables, and additional functionalities. An empirical example of fish farms illustrates the advantages of the introduced methodologies and the functionalities of the R-package.

## 1. Introduction

Data Envelopment Analysis (DEA) has been introduced to estimate relative efficiencies among decision-making units (DMUs) [1]. The estimation of DEA efficiency scores is based on a frontier constructed from the best-performing DMUs in regard to their usage of input and production of output. Inefficient DMUs are evaluated according to their respective distance to the frontier. The distance can be measured in different directions, as specified in the orientation of the model. This can, for example, be input-orientated, output-orientated, or specified with a directional distance function. The hyperbolic estimation used in this paper is a specific form of the directional distance functions that has a simultaneous input–output orientation and follows a non-linear direction to the frontier [2]. This nonlinearity is unique to the hyperbolic estimation whereas all other orientations follow a linear direction to the frontier.

The nonlinearity of the hyperbolic DEA estimation is particularly useful for between-group comparisons using different frontiers. When comparing the DMUs from one group to the frontier of another group, DMUs can be located outside the production possibility set (PPS) they are compared with e.g. in the estimation of the Malmquist index or super-efficiency scores [3,4]. That can lead to infeasibilities and therefore undefined efficiency estimations in linear DEA estimations [5–7]. This issue can be avoided when the DEA estimation is based on the assumption of constant returns to scale (CRS). However, if the underlying technology is defined by variable returns to scale (VRS), non-decreasing returns to scale (NDRS), or non-increasing returns to scale (NIRS), the DEA estimation may not ensure well-defined efficiency measures when doing between-group comparisons. A proposed solution is the estimation of hyperbolic efficiencies, where the projection to the frontier follows a hyperbolic curve [3,4,8]. If the observed data is solely positive, it is always possible to identify a benchmark for DMUs outside a PPS. The hyperbolic projection with concurrent adjustments of inputs and outputs will, by definition, lead to an intersection with any frontier [3]. Hence, it avoids issues of infeasibility and undefined efficiency scores in group comparisons when applying the DEA methodology.

However, it is not possible to represent the hyperbolic optimization problem in the respective mutual spaces referred to as envelopment (primal) and multiplier (dual) formulations [1,9]. Typically, weight restrictions, that adjust the PPS, are defined in the multiplier model. The hyperbolic estimation is only defined in the envelopment formulation, which affects the inclusion of weight restrictions. A unique approach

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to incorporate weight restrictions from the multiplier model via trade-offs in the envelopment model is provided by Podinovski [9]. This formulation of weight restrictions in the envelopment space can be used for the hyperbolic DEA estimation. In the present paper, we build on the theoretical framework by Podinovski [9] and introduce the mathematical formulation to include weight restrictions in the hyperbolic estimation. Furthermore, there is limited consideration in the literature of non-discretionary variables in the hyperbolic DEA method which enable adjustments in the model's orientation. Following the formulation by Banker and Morey [10], we introduce non-discretionary variables in the hyperbolic estimation. In a final step we combine our formulation with the generalized distance function introduced by Chavas and Cox [11] that allows further adjustments in the orientation of the model.

The present paper increases the application possibilities of the hyperbolic DEA estimation. It introduces weight restrictions and non-discretionary variables to increase the discriminatory power of efficiency estimations, avoid corner solutions as well as efficiencies by default, incorporate possible substitution effects, and enable the inclusion of prior knowledge.

For an operationalization of the introduced methodology, the non-linear form of the hyperbolic estimation is challenging to optimize. This affects the computational complexity and makes it more complicated to implement the estimation in statistical software which has restrained its popularity [3,12]. To address this challenge and make hyperbolic estimations more accessible, a function has been developed in R which will be used in this paper. It offers, among other things, the newly introduced methodology to include weight restrictions and non-discretionary variables, which, to the best of our knowledge, are currently unavailable in the statistical software R. Additional functionalities are slack estimations, super efficiencies, and external referencing. The function will be made available with this article as an R-package called *hyperbolicDEA*. It allows reproducibility as well as the application for others where hyperbolic estimations are relevant.

To showcase the methodology of hyperbolic efficiencies with the newly introduced adjustments, an illustrative application is provided with data made available alongside the R-package. It focuses on fish farms with the distinction between two species: *pangas* and *tilapia*. Hence, these species will be used as empirical examples for between-group comparisons using VRS, illustrating one of the main advantages of applying the hyperbolic efficiency measure as it avoids infeasibilities. Furthermore, it represents a general example of the application of the *hyperbolicDEA* R-package and the different functionalities it facilitates.

The rest of the paper is structured as follows. Section 2 introduces the hyperbolic estimation with the inclusion of weight restrictions and non-discretionary variables. Section 3 applies the mathematical formulation to the empirical example of aquaculture in Bangladesh. Section 4 concludes.

## 2. Methodology

For the mathematical formulation of the problem, consider the following notation with $N$ observed production units (also referred to as DMUs), each using $M$ inputs to produce $S$ outputs. Let $(x_k^n; y_r^n)$; $n = 1; \dots; N$; $k = 1 \dots; M$; $r = 1; \dots; S$ denote the observed values for inputs ($x$) and outputs ($y$) with the $n$'th DMU, $k$'th input, and $r$'th output. It is generally assumed in DEA that all observed input- and output values are nonnegative, i.e. $x_k^n \in R_0^+$ and $y_r^n \in R_0^+$. Furthermore, each DMU does, at least, use one input and produce one output. Hence, $\forall n \in N; \exists k \in M$ such that $x_k^n > 0$ and $\forall n \in N; \exists r \in S$ such that $y_r^n > 0$. Additionally, it is assumed that each output $r$, has at least one positive observed value ($\forall r \in S; \exists n \in N$ such that $y_r^n > 0$). Similarly, the input matrix is assumed to have at least one positive value for each input $k$ ($\forall k \in M; \exists n \in N$ such that $x_k^n > 0$) [9].

When DMUs are compared to a frontier constructed disregarding their own information about inputs and outputs, it is possible that the DMUs are outside the PPS. A linear orientation might then not lead to an intersection with the frontier, resulting in undefined efficiency values [3,6]. This issue can be addressed with the non-linear hyperbolic efficiency estimation. Infeasabilities can be fully circumvented, if the nonnegativity assumption on all observed input- and output values is further restricted to $x_k^n \in R^+$ and $y_r^n \in R^+$. Relaxing this assumption does not imply infeasibilities, however, makes it theoretically possible as shown by Johnson and McGinnis [3]. In general, hyperbolic estimations can mitigate issues of infeasibility and therefore enhance the precision of between-group comparisons. Consequently, it yields substantial improvements for DEA applications to further develop hyperbolic estimation techniques, as done in the present paper.

### 2.1. Weight restrictions

The standard input-oriented DEA efficiency score for some production unit $(x_k^0; y_r^0)$, assuming constant returns to scale, can be estimated using the following envelopment form of the linear programming (LP) problem [1].

$$\begin{aligned} &\min \\ &\text{s.t.:} \\ &\sum_{n=1}^{N} {}^{n}x_k^n \le x_k^0; && k = 1; \dots; M \\ &\sum_{n=1}^{N} {}^{n}y_r^n \ge y_r^0; && r = 1; \dots; S \\ &{}^{n} \ge 0; && n = 1; \dots; N: \end{aligned} \tag{1}$$

Since all LP models have a dual formulation, model (1) can be formulated in the multiplier space as follows [13]:

$$\begin{aligned} &\max \sum_{r=1}^{S} u_r y_r^0; \\ &\text{s.t.:} \\ &\sum_{k=1}^{M} v_k x_k^0 = 1; \\ &\sum_{r=1}^{S} u_r y_r^n - \sum_{k=1}^{M} v_k x_k^n \le 0; && n = 1; \dots; N \\ &u_r; v_k \ge 0; \end{aligned} \tag{2}$$

where the $u_r$'s and $v_k$'s are the respective output and input multipliers (dual variables) for each output $r$ and each input $k$. The investigated DMU $(x_k^0; y_r^0)$ adjusts the decision variables $u_k$ and $v_r$ to maximize its efficiency score.

Now, weight restrictions can be constructed with regard to the dual variables by adding additional constraints. For each weight restriction, values $a_r$ and $b_k$ are assigned to output and input dual variables, respectively, which will result in an expansion of the PPS [14]. The nature of weight restrictions can be subdivided into two main categories which are homogeneous and non-homogeneous weight restrictions. The former approach utilizes constraints on the ratios of the dual variables, while the latter approach employs absolute values as constraints [15]. Homogeneous weight restrictions can be further subdivided into linked restrictions which are between inputs and outputs and unlinked restrictions which are within either inputs or outputs[4] [14,15]. In this paper, we consider homogeneous weight restrictions that can be both linked and unlinked, and are expressed using an inequality constraint. It is possible to incorporate multiple restrictions each represented with an additional constraint.

Assuming $j = 1; \dots; T$ weight restrictions, the values defining the restrictions are represented by the matrices $a_r^j$ and $b_k^j$ for outputs

[4] In the literature linked restrictions are also called Assurance Regions of Type II and unlinked restrictions are Assurance Regions of Type I [14,15].

and inputs, respectively. Each of the T homogeneous linear weight restrictions can be written in the standard form [16]:

$$\sum_{r=1}^{S} a_r^j u_r - \sum_{k=1}^{M} b_k^j v_k \; f \; 0 \qquad j = 1; \ldots; T: \tag{3}$$

The weight restrictions in Eq. (3) can be integrated into model (2), resulting in model (4), that is the conventional integration of weight restrictions in the dual space in DEA.

$$\begin{aligned}
&\max \sum_{r=1}^{S} u_r y_r^0; \\
&\text{s.t:} \\
&\sum_{k=1}^{M} v_k x_k^0 = 1; \\
&\sum_{r=1}^{S} u_r y_r^n - \sum_{k=1}^{M} v_k x_k^n \; f \; 0; \qquad n = 1; \ldots; N \\
&\sum_{r=1}^{S} a_r^j u_r - \sum_{k=1}^{M} b_k^j v_k \; f \; 0 \qquad j = 1; \ldots; T \\
&u_r; v_k \; g \; 0:
\end{aligned} \tag{4}$$

Model (4) can now be reformulated into the envelopment DEA model. This leads to the following input-oriented CRS DEA model with weight restrictions in the envelopment formulation [14].

$$\begin{aligned}
&\min \\
&\text{s.t:} \\
&\sum_{n=1}^{N} {}^{n} x_k^n + \sum_{j=1}^{T} {}^{j} b_k^j \; f \; x_k^0; \qquad k = 1; \ldots; M \\
&\sum_{n=1}^{N} {}^{n} y_r^n + \sum_{j=1}^{T} {}^{j} a_r^j \; g \; y_r^0; \qquad r = 1; \ldots; S \\
&{}^{n}; \; {}^{j} \; g \; 0; \qquad \forall n \in N; \; j \in T:
\end{aligned} \tag{5}$$

The weight restrictions are now incorporated in the input and output constraints and optimized by adjustments of the respective decision variable, $^{j}$, for each weight restriction j. If decision variable $^{j}$ is positive, the weight restriction j is binding and therefore affects the efficiency score of the investigated DMU. The assigned values $a_r$ and $b_k$ for each weight restriction j follow technological thinking in the envelopment space and can therefore be interpreted as trade-offs [9].

Following these formulations by Podinovski [9,14] shown in model (5), the present paper modifies the efficiency estimation to a hyperbolic formulation as introduced by Färe et al. [2]. The subsequent model assumes CRS, however, later in the mathematical formulation this will be relaxed in model (9).

$$\begin{aligned}
&\min \\
&\text{s.t:} \\
&\sum_{n=1}^{N} {}^{n} x_k^n + \sum_{j=1}^{T} {}^{j} b_k^j \; f \; x_k^0; \qquad k = 1; \ldots; M \\
&\sum_{n=1}^{N} {}^{n} y_r^n + \sum_{j=1}^{T} {}^{j} a_r^j \; g \; \tfrac{1}{-} y_r^0; \qquad r = 1; \ldots; S \\
&{}^{n}; \; {}^{j} \; g \; 0; \qquad \forall n \in N; \; j \in T:
\end{aligned} \tag{6}$$

It allows a simultaneous and proportional adjustment in inputs and outputs. However, the fraction in the output constraints makes this a non-linear problem which means that it does not have a dual formulation. We suggest that homogeneous weight restrictions can nevertheless be included in hyperbolic efficiency estimations by following the trade-off approach in the envelopment formulation (model (5)) instead of a value-based inclusion in the dual space (model (4)).

T linear homogeneous weight restrictions in the standard form (Eq. (3)) can be incorporated in the estimation of the *hyperbolicDEA* package by defining a $T \times (S + M)$-matrix as shown in Eq. (7). Each row represents a linear homogeneous weight restriction where $a_r^j$ assigns the trade-off values to outputs and $b_k^j$ to inputs.

$$WR = \begin{bmatrix} a_1^1 & \cdots & a_S^1 & b_1^1 & \cdots & b_M^1 \\ \vdots & \ddots & & & \iddots & \\ \vdots & & & & & \vdots \\ \vdots & \iddots & & & \ddots & \\ a_1^T & \cdots & a_S^T & b_1^T & \cdots & b_M^T \end{bmatrix} \tag{7}$$

Each weight restriction can be linked between inputs and outputs but also stay within inputs and outputs, respectively. The possibility to include weight restrictions allows adjustments to the PPS and makes the hyperbolic estimation more flexible.

### 2.2. Non-discretionary variables and generalized distance function

Non-discretionary variables are inputs and outputs that are not controllable by the DMUs, e.g. the number of competitors, market demand, or environmental variables such as temperature, precipitation, and soil quality [17]. Incorporating non-discretionary variables in the hyperbolic estimation is particularly important as all inputs and outputs are otherwise assumed to be endogenous [2]. We are following the methodology by Banker and Morey [10] to introduce non-discretionary variables in the hyperbolic estimation. Therefore, the inputs and outputs are categorized into sets that define discretionary and non-discretionary variables. With $k_{ND} \cup k_D = 1 \ldots; M;\; k_{ND} \cap k_D = \emptyset$ and $r_{ND} \cup r_D = 1; \ldots; S;\; r_{ND} \cap r_D = \emptyset$ where $r_D$ and $k_D$ are discretionary output and inputs variables, respectively, and $r_{ND}$ and $k_{ND}$ are non-discretionary output and input variables that cannot be adjusted [10]. The hyperbolic estimation with non-discretionary variables in inputs and outputs can then be expressed as follows:

$$\begin{aligned}
&\min \\
&\text{s.t:} \\
&\sum_{n=1}^{N} {}^{n} x_k^n \; f \; x_k^0; && k \in k_D \\
&\sum_{n=1}^{N} {}^{n} x_k^n \; f \; x_k^0; && k \in k_{ND} \\
&\sum_{n=1}^{N} {}^{n} y_r^n \; g \; \tfrac{1}{-} y_r^0; && r \in r_D \\
&\sum_{n=1}^{N} {}^{n} y_r^n \; g \; y_r^0; && r \in r_{ND} \\
&{}^{n} \; g \; 0; && n = 1; \ldots; N:
\end{aligned} \tag{8}$$

The non-discretionary variables are incorporated using constraints where the is discarded. Therefore, an inefficient DMU is not required to increase a non-discretionary output variable or decrease a non-discretionary input variable to reach the frontier. Similarly, for DMUs outside the PPS, non-discretionary variables are fixed and cannot be adjusted to reach the frontier. In the *hyperbolicDEA* R-package, non-discretionary variables can be incorporated with a vector that contains the respective column indices of the non-discretionary variables in the input and output matrix.

As a final step in the mathematical formulation, we combine weight restrictions and non-discretionary variables with the generalized distance function introduced by Chavas and Cox [11]. Hence, the orientation of the model can be adjusted with the parameter that allows general shifts towards an input or output orientation. The values of are defined between $0 \; f \; \; f \; 1$ and in its extreme values it contains a linear output orientation ( $= 1$) as well as input orientation ( $= 0$). If the value is set to 0.5 the model follows the *classical* hyperbolic estimation. This parameter allows a versatile range of orientations

within the hyperbolic framework and makes it possible to include value judgments regarding the adjustability of inputs and outputs.

We combine the non-discretionary variables from model (8) and the generalized distance function with the weight restrictions from model (6) to provide the most flexible form that facilitates important adjustments to solve real-life problems while properly reflecting the production process. Furthermore, we allow for VRS, NDRS, NIRS besides CRS. This results in the following general and most comprehensive formulation:

$$
\begin{aligned}
&\min \\
&\text{s.t.:} \\
&\sum_{n=1} {}^{n}x_k^n + \sum_{j=1} {}^{j}b_k^j \, f^{1-} x_k^0; && k \in k_D \qquad (9)\\
&\sum_{n=1} {}^{n}x_k^n + \sum_{j=1} {}^{j}b_k^j \, f\, x_k^0; && k \in k_{ND} \\
&\sum_{n=1} {}^{n}y_r^n + \sum_{j=1} {}^{j}a_r^j \, g\, \frac{1}{-} y_r^0; && r \in r_D \\
&\sum_{n=1} {}^{n}y_r^n + \sum_{j=1} {}^{j}a_r^j \, g\, y_r^0; && r \in r_{ND} \\
&{}^{j} \, g\, 0; && \forall j \in T;
\end{aligned}
$$

with $_3$ (CRS) $= R_0^+$; (VRS) $= \{ \in R_0^+ \delta^{\,3} {}^{N}_{n=1} {}^{n} = 1\}$; (NDRS) $= \{ \in R_0^+ \delta {}^{N}_{n=1} {}^{n} \, g\, 1\}$; (NIRS) $= \{ \in R_0^+ \delta {}^{N}_{n=1} {}^{n} \, f\, 1\}$. The corresponding input efficiency score is $^{1-}$ and the output efficiency score is .

Model (9) furthermore showcases the implementation of our approach in the statistical software R. We additionally allow for the investigated DMU $(x_k^0; y_r^0)$ to be not in the PPS, to enable external referencing and the estimation of super-efficiency scores. The method applies a hyperbolic formulation that yields a non-radial curve for the projection of a DMU onto the efficiency frontier [2]. This results in increased complexity when solving the optimization problem. To address this problem, we have developed a custom package that uses a sequential quadratic programming algorithm. This algorithm can handle nonlinear constraints and determine the optimal minimum value of the objective function.

Nevertheless, due to the nonlinearity, there exists no deterministic solution to the problem and there is only a likelihood of identifying the global minimum [18]. With increasing computational power and the development of non-linear solvers, the likelihood will however converge toward one. The *hyperbolicDEA* R-package uses these advancements by enabling parallel computation and a plug-in mechanism for new algorithms. The function we have developed offers an effective solution to the challenges associated with hyperbolic DEA estimations and is the first R-package to facilitate the inclusion of weight restrictions and non-discretionary variables in hyperbolic efficiency estimations. Another approach to solving hyperbolic DEA estimations is the reformulation into a linear programming problem following Färe et al. [19] or Halická and Trnovská [20]. However, following the general notation of hyperbolic estimations shown in model (9) makes the mathematical formulation and programming solution more comprehensible and facilitates easier modifications and advancements.

## 3. Illustrative example

To apply the introduced methodology, the economic efficiency of a sample of *pangas* and *tilapia* fish farms in Bangladesh is estimated. Furthermore, to highlight the advantage of using a hyperbolic efficiency estimation, the difference between the two respective frontiers is investigated.

**Table 1**
Descriptive statistics.

| Statistic | Median | Max | Min | St. Dev. |
|---|---|---|---|---|
| **Tilapia** (N = 278) | | | | |
| Revenue | 1.178 | 24.091 | 0.129 | 3.226 |
| Labor cost | 0.065 | 1.200 | 0.009 | 0.156 |
| Feed cost | 0.593 | 16.940 | 0.020 | 2.002 |
| Fingerling cost | 0.074 | 3.056 | 0.007 | 0.354 |
| Other operating cost | 0.042 | 0.659 | 0.005 | 0.091 |
| **Pangas** (N = 298) | | | | |
| Revenue | 2.332 | 58.093 | 0.249 | 6.266 |
| Labor cost | 0.101 | 1.400 | 0.010 | 0.252 |
| Feed cost | 1.449 | 37.921 | 0.108 | 3.910 |
| Fingerling cost | 0.145 | 3.425 | 0.007 | 0.498 |
| Other operating cost | 0.060 | 2.140 | 0.008 | 0.217 |

Note: All variables are measured in million BDT.

### 3.1. Data

The data for the application has been collected in Bangladesh and contains 298 observations of *pangas* and 278 observations of *tilapia* fish farms. To capture the production process four input variables are identified. The first input is labor costs containing the expenses for man-days in a year. The variable also accounts for family labor. The second input is feed cost, composed of formulated feed, fertilizer, and homemade feed. As a third input, the expenses for fingerling are included and the fourth input contains other operating costs (including fixed costs) with information about the expenses for energy use, water cleaning, pond maintenance, harvesting cost, land rental, fishing nets, and asset depreciation [21,22].

The output of the production process is the market value of the produced fish (revenue) [22]. All variables are aggregated on a yearly basis, observed in 2018 on farm level, and measured in Bangladeshi Taka (BDT).

Table 1 shows descriptive statistics for the selected output and input variables. Generally, it is noticeable that *pangas* farms work on a larger scale as all median and max values exceed the values for *tilapia* farms. Important to note is that fish farming in Bangladesh generally involves polyculture, however, the current data set is of farms that focus on producing one of the species.

### 3.2. Model specification

The subsequent analysis applies a hyperbolic DEA estimation with the parameter set to 0.5. Hence, we assume equally proportioned adjustments in inputs and outputs to reach the frontier. Due to the fact that we have costs and revenue in our input and output the estimated efficiency is the profitability efficiency [23–26]. Given that we cannot detach the information on inputs and outputs to quantities and prices, it is not possible to estimate technical and allocative efficiency, respectively. More comprehensive work on the theoretical foundations of profitability efficiencies has been done by Pastor et al. [23] and Barbero and Zofío [27]. An additional advantage of the hyperbolic orientation is that it avoids all infeasibilities since our input and output matrices are positive, i.e. fulfill $x_k^n \in R^+$ and $y_r^n \in R^+$.

With regards to the returns to scale, we estimate scale efficiencies as the ratio between the efficiency scores with CRS and VRS in a hyperbolic model that does not include weight restrictions or non-discretionary variables. As we have no information on prices and quantities, differences in the profitability efficiency in VRS and CRS can also occur due to price differences. Hence, the scale-efficiency should be interpreted with caution, however, can still indicate a tendency for the returns to scale assumption. Fig. 1 shows that the scale-efficient farms are not randomly distributed along the generated revenue. On average the scale efficiency for *tilapia* peaks at a revenue of 2 million BDT whereas the scale efficiency for *pangas* peaks at a revenue of 3